Question 1: Order Form

Assumptions:
The parts of same types are stored in the same cage, with same cage code.
Customer number is unique for each customers.
Part number is unique for each parts, but not unique by unit-level.
Each part has a fixed unit price, belongs to a single part type, and is stored in a single cage code.
An employee processes the order at a given order date and order time for a customer.

1NF:
By establishing a composite primary keys (orderId and partNumber), we ensure that each attribute contains atomic values.

unnormalized_order
orderId
partNumber
customerNumber
customerName
customerType
orderDate
orderTime
employeeName
partName
partType
cageCode
quantityOrdered
unitPrice

2NF:
orderDate, orderTime, customerNumber, customerName, customerType, employeeName depends on orderId not partNumber.
partName, partType, cageCode, and unitPrice depends on partNumber not orderId.
quantityOrdered depends on both orderId and partNumber.

orders
orderId
orderDate
orderTime
customerNumber
customerName
customerType
employeeName

parts
partNumber
partName
partType
cageCode
unitPrice

order_line
orderId
partNumber
quantityOrdered

3NF:
In orders table, customerName and customerType depend on customerNumber, so we separate customers table.
In parts table, we assume that partName, partType, cageCode, and unitPrice directly depend on partNumber.

orders
orderId
orderDate
orderTime
employeeName
customerNumber

parts
partNumber
partName
partType
cageCode
unitPrice

order_line
orderId
partNumber
quantityOrdered

customers
customerNumber
customerName
customerType

Question 2: Therapist Appointments

Assumptions:
A therapist works at a single branch on any given date, but can work at different branches on different days.
staffNo, patNo, and branchNo are unique.
An appointment is uniquely scheduled for a specific patient at a specific date and time.
An appointment with a therapist takes place at the therapist's branch for the day.
A therapist cannot conduct more than one appointment at the same date and time.

1NF:
patNo, appointmentDate, and appointmentTime uniquely identifies the appointment, so establishing them as a composite primary key creates atomic values.

appointment
patNo
appointmentDate
appointmentTime
staffNo
therapistName
patName
branchNo

2NF:
patName depends on patNo.
staffNo, therapistName, and branchNo depends on patNo, appointmentDate, and appointmentTime.

patient
patNo
patName

appointment
patNo
appointmentDate
appointmentTime
staffNo
therapistName
branchNo

3NF:
therapistName depends directly on staffNo.
branchNo depends directly on staffNo and appointmentDate.

patient
patNo
patName

appointment
patNo
appointmentDate
appointmentTime
staffNo

staff
staffNo
therapistName

staff_branch_schedule
staffNo
appointmentDate
branchNo

Question 3: Event Staff

Assumptions:
eNo, contractNo, and eventNo are unique.
There are multiple employees in multiple events.
An employee can sign in multiple contracts, and multiple employees and sign in the same contract.
An event can be held in one location.
Each contract applies only to one event, and there can be multiple contracts in one event.

1NF:
Establishing a composite primary key of eNo and contractNo creates atomic values.

employee_contract_hours
eNo
contractNo
hours
eName
eventNo
eventLoc

2NF:
hours depends on both eNo and contractNo.
eName depends on eNo.
eventNo and eventLoc depend on contractNo.

employee_contract_hours
eNo
contractNo
hours

employees
eNo
eName

contracts
contractNo
eventNo
eventLoc

3NF:
eventLoc directly depends on eventNo.

employee_contract_hours
eNo
contractNo
hours

employees
eNo
eName

contracts
contractNo
eventNo

events
eventNo
eventLoc